

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{144} \frac{1}{m_e^2} s_\gamma^2 (108 c_\gamma^2 \overline{y_e}^{r i 3} \overline{y_e}^{i 2 p} y_e^{r i 4} y_e^{i 1 p} + 27 \overline{y_e}^{p r} \overline{y_e}^{i 2 i 3} y_e^{p i 4} y_e^{i 1 r} (1 + s_\gamma^2) - \right. \\ & \quad 72 c_\gamma^2 \overline{y_e}^{p r} \overline{y_e}^{i 2 i 3} y_e^{p r} y_e^{i 1 i 4} + 9 y_e^{i 1 i 4} (2 \overline{y_e}^{i 2 i 3} (13 g_1^2 + 9 g_2^2) + 3 \overline{y_e}^{p i 3} \overline{y_e}^{i 2 r} y_e^{p r}) + \\ & \quad \left. \overline{y_e}^{p i 3} (27 s_\gamma^2 \overline{y_e}^{i 2 r} y_e^{p r} y_e^{i 1 i 4} + 10 g_1^2 y_e^{p i 4} \delta_{i 1 i 2}) + 14 g_1^2 \overline{y_e}^{i 2 p} y_e^{i 1 p} \delta_{i 3 i 4} \right) - \\ & \frac{1}{4} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0} [m_d^p] + \frac{27}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_d^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{5}{36} \sum_p g_1^4 \text{LF}_{4,-1} [m_d^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{8}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_d^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{3}{2} s_\gamma \frac{1}{m_e^4} \overline{y_d}^{p r} y_d^{p r} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0} [m_d^r] - \\ & \frac{1}{4} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0} [m_e^p] + \frac{9}{9} \sum_p g_1^4 \text{LF}_{3,0} [m_e^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{5}{12} \sum_p g_1^4 \text{LF}_{4,-1} [m_e^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{8}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_e^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{1}{2} s_\gamma \frac{1}{m_e^4} \overline{y_e}^{p r} \overline{y_e}^{i 2 i 3} y_e^{p r} y_e^{i 1 i 4} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0} [m_e^r] + \\ & \frac{1}{4} s_\gamma \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (2 \overline{y_e}^{p r} y_e^{p r} (s_{2\gamma} c_\gamma - s_\gamma^3) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \text{LF}_{1,0} [m_l^p] + \\ & \frac{1}{9} \sum_p g_1^4 \text{LF}_{3,0} [m_l^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \frac{5}{24} \sum_p g_1^4 \text{LF}_{4,-1} [m_l^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{4}{45} \sum_p g_1^4 \text{LF}_{5,-2} [m_l^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \frac{1}{4} s_\gamma \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \\ & \quad (6 \overline{y_d}^{p r} y_d^{p r} (-s_{2\gamma} c_\gamma + s_\gamma^3) + 6 c_\gamma \overline{y_u}^{p r} y_u^{p r} (s_{2\gamma} + s_\gamma c_\gamma) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \\ & \quad \text{LF}_{1,0} [m_q^p] + \frac{1}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_q^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{5}{72} \sum_p g_1^4 \text{LF}_{4,-1} [m_q^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{4}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_q^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0} [m_u^p] + \frac{27}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_u^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{5}{9} \sum_p g_1^4 \text{LF}_{4,-1} [m_u^p] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{32}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_u^p] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{3}{2} s_\gamma c_\gamma \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \overline{y_u}^{p r} y_u^{p r} (s_{2\gamma} + s_\gamma c_\gamma) \text{LF}_{1,0} [m_u^r] - \frac{1}{8} s_\gamma \frac{1}{m_e^4} \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \\ & \quad (3 s_{4\gamma} c_\gamma (g_1^2 + g_2^2) + s_\gamma (g_1^2 (-1 + 3 c_{2\gamma}^2) + 3 g_2^2 (-1 + c_{2\gamma}^2))) \text{LF}_{1,0} [m_\Phi] + \\ & \frac{1}{8} \frac{1}{m_e^2} (3 s_{4\gamma} \overline{y_e}^{p r} \overline{y_e}^{i 2 i 3} y_e^{p i 4} y_e^{i 1 r} + s_\gamma^2 y_e^{i 1 i 4} (-2 \overline{y_e}^{i 2 i 3} (g_1^2 + 3 g_2^2) + 3 s_\gamma^2 \overline{y_e}^{p i 3} \overline{y_e}^{i 2 r} y_e^{p r})) \\ & \quad \text{LF}_{1,1} [m_\Phi] + \frac{1}{24} s_\gamma^2 (3 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} (g_1^2 + 3 g_2^2) - 4 g_1^2 \overline{y_e}^{p i 3} y_e^{p i 4} \delta_{i 1 i 2} - \\ & \quad 4 \overline{y_e}^{i 2 p} y_e^{i 1 p} (3 c_\gamma^2 \overline{y_e}^{r i 3} y_e^{r i 4} + 2 g_1^2 \delta_{i 3 i 4})) \text{LF}_{1,2} [m_\Phi] + \\ & \frac{1}{12} s_\gamma^2 (-g_1^2 \overline{y_e}^{p i 3} y_e^{p i 4} \delta_{i 1 i 2} + \overline{y_e}^{i 2 p} y_e^{i 1 p} (-3 s_\gamma^2 \overline{y_e}^{r i 3} y_e^{r i 4} + g_1^2 \delta_{i 3 i 4})) \text{LF}_{2,1} [m_\Phi] + \\ & \frac{1}{36} g_1^2 (3 s_\gamma^2 \overline{y_e}^{p i 3} y_e^{p i 4} \delta_{i 1 i 2} + (-3 s_\gamma^2 \overline{y_e}^{i 2 p} y_e^{i 1 p} + 4 g_1^2 \delta_{i 1 i 2}) \delta_{i 3 i 4}) \text{LF}_{3,0} [m_\Phi] - \\ & \frac{5}{24} g_1^4 \text{LF}_{4,-1} [m_\Phi] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{4}{45} g_1^4 \text{LF}_{5,-2} [m_\Phi] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{9} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{1}{6} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i 1 i 2} \delta_{i 3 i 4} - \\ & \frac{8}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{1}{2} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{1,1,0} [m_1, m_e^{i 3}] - \\ & \frac{1}{4} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{2,1,-1} [m_1, m_e^{i 3}] + \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i 3}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i 3}] \delta_{i 1 i 2} \delta_{i 3 i 4} - \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i 3}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i 3}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{1}{2} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{1,1,0} [m_1, m_e^{i 4}] - \\ & \frac{1}{4} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{2,1,-1} [m_1, m_e^{i 4}] + \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i 4}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i 4}] \delta_{i 1 i 2} \delta_{i 3 i 4} - \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i 4}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i 4}] \delta_{i 1 i 2} \delta_{i 3 i 4} + \frac{1}{8} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{1,1,0} [m_1, m_l^{i 1}] - \\ & \frac{1}{16} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e}^{i 2 i 3} y_e^{i 1 i 4} \text{LF}_{2,1,-1} [m_1, m_l^{i 1}] + \frac{1}{48} g_1^4 \text{$$